

Test Diverse Users

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Build the Recommender

Without opening the old lesson, explain one key idea from Build the Recommender. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Permissions and Human Review

Without opening the old lesson, explain one key idea from Permissions and Human Review. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Prompt Experiments

Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Plan coverage for Evaluating varied cases. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Evaluating varied cases. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Evaluate independently

Given ordinary 3/3, boundary 1/2, invalid 2/2 and tie 0/1, compute overall results and prioritize issues.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

The recommender succeeds on 9/10 mystery requests and 2/5 poetry requests. Compute total and separate rates. A team reports only 11/15. Explain what that hides.

6. Solve the given case

A recommender satisfies 5/6 adventure requests and 3/6 poetry requests.

Compare and investigate.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

A recommender serves 6 of 6 easy-book requests but 1 of 4 advanced-book requests. Report both results and identify the next test priority.

My independent answer and reason

Write or draw your reasoning and evidence.

Week 34 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Use the same test record across ordinary, boundary, missing-data, and access-needs cases.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

The recommender succeeds on 9/10 mystery requests and 2/5 poetry requests. Compute total and separate rates. A team reports only 11/15. Explain what that hides.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Read denominators

A report says 8 of 10 tests pass, including 0 of 2 invalid-input cases. What is the important weakness?

My reasoning and evidence

13. W2 Plan coverage

Name five distinct categories for the chooser evaluation.

My reasoning and evidence

14. W3 Repair reporting

A learner drops two failed cases and reports 8/8. What should be restored?

My reasoning and evidence

15. W4 Evaluate independently

Given ordinary 3/3, boundary 1/2, invalid 2/2 and tie 0/1, compute overall results and prioritize issues.

My reasoning and evidence

16. W5 Protect final evidence

You fix the tie bug using the final tie case. What should the next evaluation report say?

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Test Diverse Users

A fictional recommender is tested on distinct ordinary, equality, invalid-input and tie cases.

Recorded full passes: ordinary 4/4; equality 1/3; invalid 2/3; ties 1/2.

Categories are disjoint and every case was run on the same frozen version with the same contract.

Disjoint evaluation categories

	Full passes	Cases run
Ordinary	4	4
Equality	1	3
Invalid	2	3
Ties	1	2

Read a cell using both its row and column labels.

Test Diverse Users

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Total count

Passes $4+1+2+1=8$; cases $4+3+3+2=12$.



2. Overall rate

$8/12 \approx 66.7\%$ full passes.



3. Find weak slices

Equality 1/3; invalid 2/3; ties 1/2; ordinary 4/4.



4. Prioritise repair

Inspect failed equality traces and any invalid-input safety consequence, not only the total.

CHANGE ONE THING

Someone reports only the ordinary slice.

Explain the changed result using the matching lesson.

Can these records establish performance for different user groups?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition coverage.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

Build a recommender evaluation matrix with routine, boundary, missing, invalid and tie cases, then add separately planned user-condition coverage.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional recommender is tested on distinct ordinary, equality, invalid-input and tie cases.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Inspect failed equality traces and any invalid-input safety consequence, not only the total.

1. Prioritise repair
2. Total count
3. Find weak slices
4. Overall rate

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Prioritise repair Inspect failed equality traces and any invalid-input safety consequence, not only the total. Varied tests expose behavior an ordinary demonstration misses. The overall result combines disjoint cases once each; overlapping labels would require unique-case accounting. Prioritisation also depends on consequences: a malformed input that triggers an unauthorised action can be more serious than a harmless tie-order failure. Retain each failed case and its expected output.

Practice 2: explain the deciding evidence

Changed-case practice: Someone reports only the ordinary slice. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. The claim becomes 4/4 on ordinary cases, not 100% across all tests.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The claim becomes 4/4 on ordinary cases, not 100% across all tests. Eight nonordinary cases and their failures have been omitted.

Repair a missing building block

Review Build the Recommender (ai-2/33). Goal: Implement a deterministic recommender. Short practice: Explain why B does not win after D is excluded even though it has fewer pages.. Return to this week and explain what you changed.

Review Permissions and Human Review (ai-2/31). Goal: Explain the mechanism and terminology of least privilege and tool scopes. Short practice: Which proposal is allowed, and why is an authorised object alone insufficient?. Return to this week and explain what you changed.

Review Prompt Experiments (ai-2/28). Goal: Explain the mechanism and terminology of controlled prompt experiments. Short practice: Name the case P passes and Q fails, and explain why it matters.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
